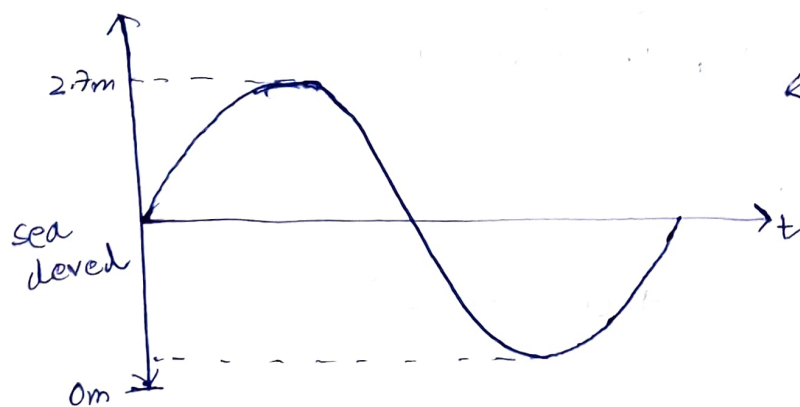


Tide Harmonics

Sea level behave like high, low, high, low

So it's like a wave



Suppose we have this

Time	Height
22:00 02:02	2.70 m
08:33	0.33 m
14:12	2.63 m
20:55	0.37 m

Our target? $\Rightarrow$ Find a mathematical function that behaves like this and use that function to estimate future tides.

But a tide is not one wave

$\rightarrow$ not a simple sin or cosine wave

It's like 3 people pushing a swing

(A)
pushes very
regularly

(B)
pushes in
different
rhythm than (A)

(C)
pushes more slowly

$\therefore$ The observed Tide is, Tide = wave 1 + wave 2 + wave 3 ...

Tidal constituents.

~~Q~~ We are using 6 tidal constituents among 38 major.

- M2 $\rightarrow$ Principal Lunar Semidiurnal
- S2 $\rightarrow$ Principal Solar Semidiurnal
- K1 $\rightarrow$ Lunisolar Diurnal
- O1 $\rightarrow$ Principle Lunar Diurnal
- M4 $\rightarrow$ Shallow water equator-Diurnal
- N2 $\rightarrow$ Larger Lunar elliptic Diurnal

From Fourier analysis we know,

$$h(t) = H_0 + \sum (F_i H_i \cos(\omega_i t + \phi_i))$$

in our case,

$$(i) \rightarrow h(t) = Z_0 + \sum_i H_i \cos(\omega_i t - \phi_i) \quad \left[\begin{array}{l} h(t) \rightarrow \text{water level} \\ \text{at time } t \end{array} \right]$$

$$\left[19 \text{ August } 00:00 = t = 0 \right] \quad \left[Z_0 = \text{average sea level} \right]$$

$$\left[H_i = \text{amplitude} \right] \quad \left(\text{How strongly particular wave affect the tide} \right)$$

$$\left[\omega_i = \text{speed / frequency of the wave} \right] \quad \left[\phi_i = \text{phase} \right]$$

$\rightarrow$ known, universal astronomical data

Simplify equation (i)

$$\text{we know } \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\therefore H_i \cos(\omega_i t - \phi_i)$$

$$= H_i \cos(\omega_i t) \cos(\phi_i) + H_i \sin(\omega_i t) \sin(\phi_i)$$

$$= a_i \cos(\omega_i t) + b_i \sin(\omega_i t)$$

$$\therefore h(t) = z_0 + \sum_i [a_i \cos(\omega_i t) + b_i \sin(\omega_i t)]$$

it looks like a

$$y = a + bx$$

$\therefore$ unknowns are a & b

So we need regression to estimate it.

Same way look at our equation.

$$h(t) = z_0 + a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) + a_2 \cos(\omega_2 t) + b_2 \sin(\omega_2 t)$$

The ' ω ' were known already.

So, we know

$$\cos(\omega_i t) \text{ or } \sin(\omega_i t)$$

we need MLRM for this.

So How should one row of data will look like?

Suppose take date 19 Aug. 2026, 2802, 2.70m

$$\therefore t = 2.033 \text{ hours}$$

$$\text{For } m_2, \quad \omega_{m_2} = 28.9841042^\circ/\text{hr}$$

$$\text{or } \omega_{m_2} = 0.5059 \text{ radian./hr}$$

$$\therefore \omega_{m_2} t = 1.029 \text{ radian}$$

$$\cos(1.029) \approx 0.516$$

$$\sin(1.029) \approx 0.857$$

one row will look like

$$[1, 0.5160, 0.857, 0.485, 0.875, \dots]$$

Design Matrix Matrix

observation	1	$\cos m_2$	$\sin m_2$	$\cos s_2$	$\sin s_2$	Height
1	1	0.516	0.857	0.485	0.875	2.70
2	1	...	...	...	...	0.33
3	1	...	...	...	...	2.63

$[x] \rightarrow$ explanatory variable

$[y] \rightarrow$ observed height

$[\beta] \rightarrow$ unknown coefficient

$$y \approx x\beta \rightarrow \text{whole model}$$

$$[\beta] =$$

$$\begin{bmatrix} z_0 \\ a_{m_2} \\ b_{m_2} \\ a_{s_2} \\ b_{s_2} \\ a_{N_2} \\ b_{N_2} \\ a_{x_1} \\ b_{x_1} \\ a_{o_1} \\ b_{o_1} \\ a_{m_4} \\ b_{m_4} \end{bmatrix}$$

(13) unknown coefficient

we have

We have only 38 observation and all are extrema
from those we have to found 13 unknown coefficient.

$$\text{error} = \text{actual height} - \text{predicted height}$$

$\therefore$ errors are $e_1, e_2, e_3, \dots$

$$\therefore \text{SSE} = \sum_j e_j^2 = \sum_j (x_j - \hat{y}_j)^2$$

So computer tries some coefficients:

Suppose

$$z_0 = 1.5$$

$$a_{m2} = -0.07$$

$$b_{m2} = 1.27$$

$\vdots$

} → from this it predicts 38 predicted heights
↓

<u>actual</u>	<u>predicted</u>	<u>error</u>
2.70	2.68	0.02
0.33	0.34	-0.01
2.63	2.64	-0.01

↓
calculate SSE from it

repeat with
new coefficient
in search of smallest (SSE)

the mathematical solution to linear least square is

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$X' = \text{transpose}$$

But for this complicated matrix it's unstable to find $\hat{\beta}$ from calculating $(X'X)^{-1}$

Derivation

$$SSE = e'e = (y - X\beta)'(y - X\beta)$$

$$SSE = (y' - \beta'X')(y - X\beta)$$

$$SSE = y'y - y'X\beta - \beta'X'y + \beta'X'X\beta$$

$$SSE = y'y - 2\beta'X'y + \beta'X'X\beta$$

Minimize SSE

$$\frac{\partial(SSE)}{\partial \beta} = -2X'y + 2X'X\beta = 0$$

$$\text{or } \hat{\beta} = (X'X)^{-1}X'y$$

Householder QR

Factor our matrix X in two special matrix

$$X = QR \quad \left[\begin{array}{l} Q = \text{orthogonal matrix} \\ R = \text{upper triangular matrix} \end{array} \right]$$

So,

$$\min \|y - X\beta\|^2$$

$$\text{or } \|y - QR\beta\|^2$$

$$\text{or } \|Q'(y - QR\beta)\|^2 = \|Q'y - Q'QR\beta\|^2 \quad [Q'Q = I]$$

$$\propto \|Q'y - RB\|^2$$

$$\Rightarrow Q'y = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}, R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

$$\therefore \left\| \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} - \begin{bmatrix} R_1 \\ 0 \end{bmatrix} B \right\|^2 = \|c_1 - R_1 B\|^2 + \|c_2\|^2$$

To minimize this, first term can be made completely '0'

by getting: $R_1 B = c_1$

Computer will back solve this for B

using this computer will solve

from all these computer predicts final 6 constituent approx.

Constituent	Amplitude(H)	Phase(θ)
M2	1.2731 m	266.66°
S2	0.6478 m	354.44°
N2	0.1799 m	145.05°
K1	0.0815 m	163.05°
O1	0.0416 m	318.48°
M4	0.0285 m	33.82°

How 'H' & ' θ ' get's predicted?

$$\begin{aligned} H \cos(\omega t - \theta) &= H \cos \theta \cos \omega t + H \sin \theta \sin \omega t \\ &= a \cos \omega t + b \sin \omega t \end{aligned}$$

$$\therefore a^2 + b^2 = H^2 \cos^2 \theta + H^2 \sin^2 \theta = H^2$$

$$\therefore H = \sqrt{a^2 + b^2}$$

again $\frac{a}{b} = \frac{H \cos g}{H \sin g}$

or $\frac{a}{b} = \tan g$

or $g = \tan^{-1}(a/b)$

or $g = \text{atan2}(-b, a)$

Computer take $\text{atan2}(y, x)$
for specific tidal program
not $\text{atan2}(x, y)$.

here a (-) sign before b
because in harmonic (tidal)
phase lags are measured
backwards. To make sure
computer angle output
rotate in correct ~~phase~~ direction
we introduce a negative sign.

So the architecture:

Observed tide data



Create sin, cosin variable



Least square regression



Estimate a and b



convert a, b



construct final tide equation.



predict future tide.

~~Why 5 models?~~

Why 5 different models?

- (i) $M2 + S2$
- (ii) $M2 + S2 + N2$
- (iii) $M2 + S2 + N2 + K1$
- (iv) $M2 + S2 + N2 + K1 + O1$
- (v) $M2 + S2 + N2 + K1 + O1 + M4$

Program tested on all 5 models and winner is (v)

We can take more models ~~but~~ with more ~~more~~ constituents but that might cause overfitting.

because memorised sample of learning the underlying pattern.

Cross validation

Given data has 10 day (38 observation)



Computer hides each day and ~~hides~~ use other data to estimate the model ~~for each day~~ ~~and~~ each hidden day. Leave one day out cross validation

Result

<u>Model</u>	<u>Fit RMSE</u>	<u>LOO-CV RMSE</u>
(i)	0.0792m	0.0869m
(ii)	0.0746m	0.0800m
(iii)	0.0704m	0.0815m
(iv)	0.0176m	0.0232m
(v)	0.0137m	0.0225m

So model (v) selected not only because it had the lowest in sample ~~rmse~~ error but it also performed best when a whole day was hidden.

RMSE = Root Mean Squared Error

High tide low tide from tide equation

$$n(t) = z_0 + \sum_i [a_i \cos(\omega_i t) + b_i \sin(\omega_i t)]$$

$$\frac{dn(t)}{dt} = \sum_i \left[a_i \frac{d}{dt} \cos(\omega_i t) + b_i \frac{d}{dt} \sin(\omega_i t) \right]$$

$$= \sum_i \left[-a_i \omega_i \sin(\omega_i t) + b_i \omega_i \cos(\omega_i t) \right]$$

$$\frac{d^2 n}{dt^2} < 0 \rightarrow \text{High tide}$$

$$\frac{d^2 n}{dt^2} > 0 \rightarrow \text{low tide}$$

in our model RMSE measure Height accuracy

not focused on time accuracy^{***} [limitation and scope of improvement]

The model display moon phase, illumination, sunrise, sunset

but not taking them into account of equation [future imp. possible]